

The WKB Approximation

The WKB (Wentzel-Kramers-Brillouin) approximation is often referred to as the semi-classical approximation to solve Schrödinger equation. Before delving in, let me clarify that the WKB approximation is not the same as taking the classical limit. Rather, it is a method to solve quantum mechanical problems where the wavelength of our particle is much smaller than other characteristic lengths in the problem. Above all else, the WKB approximation is concerned with the phase of the wavefunction as the particle propagates through a potential region that varies slowly in space. Note that as opposed to the variational method, where we had stationary solutions in mind, the WKB approximation is a method to find *dynamical* solutions of the time-dependent Schrödinger equation.

Preliminaries and Classical Limit

We start with the Schrödinger equation in three-dimensions with some generic potential $V(\mathbf{x}, t)$:

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = -\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}, t) + V(\mathbf{x}, t) \psi(\mathbf{x}, t). \quad (1)$$

As you have done in your homework, we can write the wavefunction without any loss of generality in terms of an amplitude and a phase as

$$\psi(\mathbf{x}, t) = \sqrt{\rho(\mathbf{x}, t)} e^{\frac{i}{\hbar} S(\mathbf{x}, t)}, \quad (2)$$

where $\rho(\mathbf{x}, t) = |\psi(\mathbf{x}, t)|^2$ is the probability density and $S(\mathbf{x}, t)$ is some phase. Both are real functions. Recall that with this parameterization we found the probability density current to be

$$\mathbf{J}(\mathbf{x}, t) = \frac{\hbar}{2mi} (\psi^*(\mathbf{x}, t) \nabla \psi(\mathbf{x}, t) - \psi(\mathbf{x}, t) \nabla \psi^*(\mathbf{x}, t)) = \rho(\mathbf{x}, t) \frac{\nabla S(\mathbf{x}, t)}{m}. \quad (3)$$

The interpretation was that $\nabla S(\mathbf{x}, t)$ is analogous to the momentum (or $\nabla S(\mathbf{x}, t)/m$ is analogous to the velocity, $p = mv$) of a classical particle at position $\mathbf{x}$ and time t . It is in this sense that we will think of the phase of the wavefunction as encoding information about the dynamics of our particle.

To see this, take a free particle traveling through space with a wavefunction

$$\psi(\mathbf{x}, t) = C e^{\frac{i}{\hbar} (\mathbf{p} \cdot \mathbf{x} - Et)}, \quad (4)$$

where C is a normalization constant, $\mathbf{p}$ is the particle three-momentum and $E = \frac{\mathbf{p}^2}{2m}$ is its energy. Clearly, for this simplest case,

$$S(\mathbf{x}, t) = \mathbf{p} \cdot \mathbf{x} - Et \quad \implies \quad \begin{cases} \nabla S(\mathbf{x}, t) = \mathbf{p}, \\ \frac{\partial S(\mathbf{x}, t)}{\partial t} = -E. \end{cases} \quad (5)$$

As expected, we recover the momentum, but what is more, we also learn that the time derivative is just $-E$. Both of these equalities are consistent with the classical action in classical mechanics. Note that for this single particle case $\rho(\mathbf{x}, t) = |C|^2$, which is subtle since a free particle is not normalizable in all space, but that is a whole separate issue.

What is important here is that in this example, the phase depends trivially on position and time thanks to the absence of a potential. Once you turn on a potential $V(\mathbf{x}, t)$, the phase $S(\mathbf{x}, t)$ will in general be more complicated and could display more interesting dynamics than the motion of our free particle plane wave. It is precisely this dynamical problem that the WKB approximation aims to solve.

Plugging the form of Eq. (2) into Schrödinger's equation, we find

$$i\hbar \left(\frac{1}{2\sqrt{\rho}} \frac{\partial \rho}{\partial t} + \frac{i}{\hbar} \sqrt{\rho} \frac{\partial S}{\partial t} \right) e^{\frac{i}{\hbar} S} = -\frac{\hbar^2}{2m} \left[\nabla^2 (\sqrt{\rho} e^{\frac{i}{\hbar} S}) \right] + V \sqrt{\rho} e^{\frac{i}{\hbar} S}, \quad (6)$$

where I dropped my arguments for simplicity. The laplacian term is just $\nabla^2 (\sqrt{\rho} e^{\frac{i}{\hbar} S}) = e^{\frac{i}{\hbar} S} [\nabla^2 \sqrt{\rho} + \frac{2i}{\hbar} \nabla \sqrt{\rho} \cdot \nabla S + \frac{i}{\hbar} \sqrt{\rho} \nabla^2 S - \frac{1}{\hbar^2} \sqrt{\rho} (\nabla S)^2]$. Dividing both sides by $e^{\frac{i}{\hbar} S}$ and separating real and imaginary parts (we can do this because we know that both S and ρ are real functions), you should find:

$$\frac{\partial S}{\partial t} + \frac{(\nabla S)^2}{2m} + V - \frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}} = 0, \quad (7)$$

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\rho \frac{\nabla S}{m} \right) = \frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0. \quad (8)$$

The second equation is just the continuity equation for probability density that we have already seen (this is significant! the conservation of probability is encoded in the imaginary part of the Schrödinger equation). The first equation is the one that concerns us and the WKB method.



Connection to classical mechanics: Eq. (7) looks very similar to the Hamilton-Jacobi equation from classical mechanics, which reads (see Eq. 2.8.18 of Shankar's book):

$$\frac{\partial S}{\partial t} + H(\mathbf{x}, \nabla S, t) = 0, \quad (9)$$

where $H(\mathbf{x}, \mathbf{p}, t)$ is the Hamiltonian of the system expressed in terms of coordinates $\mathbf{x}$ and momenta $\mathbf{p}$. For a single non-relativistic particle in a potential $V(\mathbf{x}, t)$, the Hamiltonian is $H = \frac{\mathbf{p}^2}{2m} + V(\mathbf{x}, t)$, so using our analogy between momentum and the divergence of the phase, the Hamilton-Jacobi equation becomes

$$\frac{\partial S}{\partial t} + \frac{(\nabla S)^2}{2m} + V(\mathbf{x}, t) = 0. \quad (10)$$

It is intriguing that following this reasoning, we see that Eq. (7) reduces to the Hamilton-Jacobi equation only in the limit $\hbar \rightarrow 0$ (classical limit), where the last term vanishes. This is one way to see how classical behavior emerges from quantum mechanics in the appropriate limit.

First Hints of Semi-Classical Behavior

As we will see below, the WKB approximation is technically an expansion in $\hbar$. Well, one obvious feature of the last term in Eq. (7) is that it comes in at order $\hbar^2$, and so it should be small when the WKB method is useful. This last term is sometimes called the quantum potential:

$$Q(\mathbf{x}, t) = -\frac{\hbar^2}{2m} \frac{\nabla^2 \sqrt{\rho(\mathbf{x}, t)}}{\sqrt{\rho(\mathbf{x}, t)}}. \quad (11)$$

Physically, it encodes quantum effects in “wave mechanics”, and it is responsible for quantum phenomena such as tunneling and interference. While called a potential, this is very different from $V(\mathbf{x}, t)$ as it depends on our state $\psi(\mathbf{x}, t)$ itself (through $\rho(\mathbf{x}, t)$). Without it, we recover the Hamilton-Jacobi equation.

Let us see this more explicitly by taking the gradient of Eq. (7) and identifying $\mathbf{v} = \nabla S/m$ as before. We get a $F = ma$ type equation, with $F = -\nabla(V + Q)$:

$$m \left(\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right) = -\nabla(V + Q). \quad (12)$$

In one dimension, we can interpret the term inside the parenthesis on the left-hand side as just $d(v)/dt = \partial_t v + \frac{\partial v}{\partial x} \frac{\partial x}{\partial t} = \partial_t v + v \frac{\partial v}{\partial x}$. In the WKB approximation, the phase $S(\mathbf{x}, t)$ of the wavefunction satisfies the classical Hamilton-Jacobi equation to a good approximation, and the dynamics of the particle can be understood in terms of classical trajectories.

The WKB Approximation (In Practice)

Okay, so we understand that there is some limit where the quantum behavior is irrelevant when $\hbar \rightarrow 0$. But in the way we wrote it above, the WKB approximation is not very useful. It still does not make our jobs easier to solve Schrödinger's equation and we never really found any expansion parameter ($\hbar$ is a constant after all!)

To see the usefulness of the WKB approximation, let us consider the time-independent Schrödinger equation in three-dimensions with some generic potential $V(\mathbf{x})$ and energy eigenvalue E :

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}) + V(\mathbf{x}) \psi(\mathbf{x}) = E \psi(\mathbf{x}). \quad (13)$$

Note that the fact that we are seeing the energy eigenvalue E means that our wavefunction $\psi(\mathbf{x})$ is an energy eigenstate (that is, we have used $\hat{H}|\psi\rangle = E|\psi\rangle$ to arrive at the equation above). Because we will be thinking about the dynamics of a particle with a given momentum, it is also convenient to re-write Schrödinger's equation in terms of the wavevector $\mathbf{k}$, defined as $\mathbf{p} = \hbar \mathbf{k}$. Rearranging, we have

$$\hat{\mathbf{p}}^2 \psi(\mathbf{x}) = 2m(E - V(\mathbf{x})) \psi(\mathbf{x}) \quad \leftrightarrow \quad -\hbar^2 \nabla^2 \psi(\mathbf{x}) = 2m(E - V(\mathbf{x})) \psi(\mathbf{x}), \quad (14)$$

where again I emphasize that $\hat{\mathbf{p}} = -i\hbar \nabla$ is the momentum operator that acts on functions as opposed to $\hat{p}|\psi\rangle$ which is the equivalent operator acting on kets in Hilbert space. The fact that I use the same notation for both is a common abuse of notation, but which operator I am referring to should be clear from the context.

Now, instead of working with Eq. (2), let us make a similar, but conceptually distinct ansatz for the energy eigenstate wavefunction:

$$\psi(\mathbf{x}) = e^{\frac{i}{\hbar} S(\mathbf{x})}, \quad (\text{with energy } E), \quad (15)$$

where $S(\mathbf{x})$ is now a complex function rather than a real one. Note the lack of a prefactor to the exponential, which is now not necessarily a complex exponential(!). This form is as general as Eq. (2).¹ Plugging this form into the time-independent Schrödinger equation in Eq. (14) (where now we have the extra assumption of this being an energy eigenstate and only have to worry about the phase $S(\mathbf{x})$), we find

$$-\hbar^2 \nabla^2 \left(e^{\frac{i}{\hbar} S(\mathbf{x})} \right) = i\hbar \nabla^2 S(\mathbf{x}) - (\nabla S(\mathbf{x}))^2 = 2m(E - V(\mathbf{x})). \quad (16)$$

Now here comes the key step in the WKB approximation: we expand $S(\mathbf{x})$ in powers of $\hbar$ as

$$S(\mathbf{x}) = S_0(\mathbf{x}) + \hbar S_1(\mathbf{x}) + \hbar^2 S_2(\mathbf{x}) + \dots, \quad (17)$$

where $S_0(\mathbf{x})$ is the leading order term, $S_1(\mathbf{x})$ is the first order correction, and so on.

¹We can always write $S(\mathbf{x}) = R(\mathbf{x}) + iI(\mathbf{x})$ and then $\psi(\mathbf{x}) = e^{iR(\mathbf{x})/\hbar} e^{-I(\mathbf{x})/\hbar}$, which is equivalent to Eq. (2) as long as we interpret $\sqrt{\rho(\mathbf{x})} = e^{-I(\mathbf{x})/\hbar}$.



This should make you uncomfortable for two reasons: 1) $\hbar$ is a *dimensionful* constant in nature, so what good can it do as an expansion parameter? In natural units, $\hbar = 1$! 2) how can we be sure that this series converges without already knowing the full solution?

The first concern is resolved by noting that what really matters is the ratio of $\hbar$ to some characteristic action S_c in the problem, which has the same units as $\hbar$ (action has units of energy times time, or equivalently momentum times length). Thus, in some sense, the expansion parameter is really in terms of a dimensionless ration like $\hbar/S_c$, which is dimensionless. The second concern is more subtle and we have no way of proving it in all generality. That being said, we know for a fact that taking $\hbar$ to be small recovers classical behavior.